

```

    {
        perror("Sender: mq_open(sender)");
        exit(1);
    }
    else
    {
        printf("Sender MQ %s opened\n", mq_name);
    }

    for(int i=0;i<10;i++)
    {
        sprintf(&tx_buffer[14], "%d", i);
        printf("Sender: sending message: [%s]\n",
tx_buffer);
        // send message to message queue
        if(mq_send(mqd_sender, &tx_buffer[0], strlen
(tx_buffer), 0) == (mqd_t)-1)
        {
            perror("Sender: Unable to send
message");
        }
    }

    if(mq_close(mqd_sender) == (mqd_t)-1)
    {
        perror("Sender: mq_close");
        exit(1);
    }

    printf("Sender: Exit\n");

    exit(0);
}

```

// pmq_receiver_ssmr.c

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <mqueue.h>

#define MQ_QUEUE_PERMISSIONS 0660
#define MQ_MAX_MESSAGES 10
#define MQ_MAX_MSG_SIZE 128
#define MQ_MSG_BUFFER_SIZE MQ_MAX_MSG_SIZE + 10

int main(int argc, char **argv)
{
    mqd_t mqd_receiver;
    const char *mq_name = "/mq-postman";
    struct mq_attr attr;
    char rx_buffer[MQ_MSG_BUFFER_SIZE];

```

```

    attr.mq_flags = 0;
    attr.mq_maxmsg = MQ_MAX_MESSAGES;
    attr.mq_msgsize = MQ_MAX_MSG_SIZE;
    attr.mq_curmsgs = 0;

    if((mqd_receiver = mq_open(mq_name, O_RDONLY | O_
CREAT, MQ_QUEUE_PERMISSIONS, &attr)) == (mqd_t)-1)
    {
        perror("Receiver: mq_open");
        exit(1);
    }
    else
    {
        printf("Receiver MQ %s opened\n", mq_name);
    }

    int i=9;
    while(i)
    {
        printf("Receiver: blocked until message
arrives...\n");
        // get the oldest message with highest
priority first from message queue
        if(mq_receive(mqd_receiver, rx_buffer, MQ_
MSG_BUFFER_SIZE, NULL) == (mqd_t)-1)
        {
            perror ("Receiver: mq_receive");
            exit (1);
        }
        else
        {
            printf("Receiver: message received:
[%s]\n", rx_buffer);
            i--;
        }
    }

    if (mq_close(mqd_receiver) == (mqd_t)-1)
    {
        perror ("Sender: mq_close");
        exit (1);
    }

    printf("Receiver: Exit\n");

    exit(0);
}

```

After compiling *pmq_receiver_ssmr.c*, make two copies of its executable.

Run the two receivers viz *pmq_receiver_ssmr* and *pmq_receiver_ssmr2* in two different terminals first, followed by *pmq_sender_ssmr* in the third terminal.